

Lecture 34: Nov 14

Last time

- Midterm Exam 2

Today

- Location Scale families
- Exponential families

Location and Scale families

Let Z be a continuous random variable with pdf $f(z)$. Define the class of random variables

$$X_{\mu,\sigma} = \sigma Z + \mu, \quad \mu \in \mathbb{R}, \sigma > 0$$

Then

1. $X_{\mu,\sigma}$ has pdf

$$f_{\mu,\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right)$$

- 2.

$$E(X) = \sigma E(Z) + \mu, \quad \text{Var}(X) = \sigma^2 \text{Var}(Z)$$

3. The variable $Z = X_{0,1}$ is called the *generator* and is a member of the class.

Location families and scale families

- The family of pdfs $f_{\mu,\sigma}(x)$ is called a *location-scale* family where μ is called the *location parameter*, and σ is called the *scale parameter*.
- The family of pdfs

$$f_{\mu,1}(x) = f(x - \mu)$$

with $\sigma = 1$ is called a *location* family.

- The family of pdfs

$$f_{0,\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right)$$

with $\mu = 0$ is called a *scale* family.

Example (Exponential location family) Let $f(x) = e^{-x}$, $x \geq 0$, and $f(x) = 0$, $x < 0$. To form a location family we replace x with $x - \mu$ to obtain

$$f(x|\mu) = \begin{cases} e^{-(x-\mu)} & x - \mu \geq 0 \\ 0 & x - \mu < 0 \end{cases}$$

$$= \begin{cases} e^{-(x-\mu)} & x \geq \mu \\ 0 & x < \mu \end{cases}$$

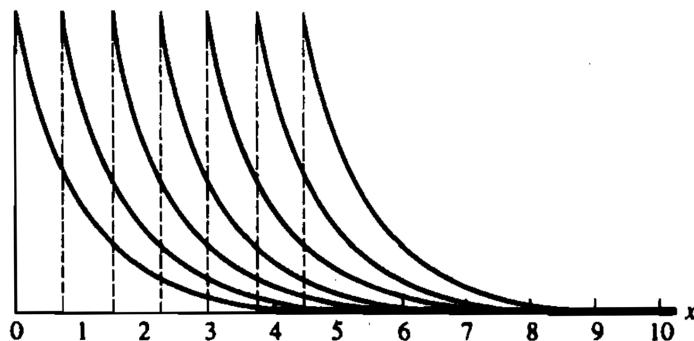


Figure 3.5.2. *Exponential location densities*

Figure 33.1: Figure 3.5.2. Exponential location densities.

As shown in the above graph, the densities are shifted. Now the positive part of the density starts at μ rather than at 0. If X measures time, then μ might be restricted to be nonnegative so that X will be positive with probability 1 for every value of μ . In this type of model, where μ denotes a bound on the range of X , μ is sometimes called a *threshold parameter*.

The effect of introducing the scale parameter σ is either to stretch ($\sigma > 1$) or to contract ($\sigma < 1$) the graph of $f(x)$ while still maintaining the same basic shape of the graph. This is illustrated in the Figure below.

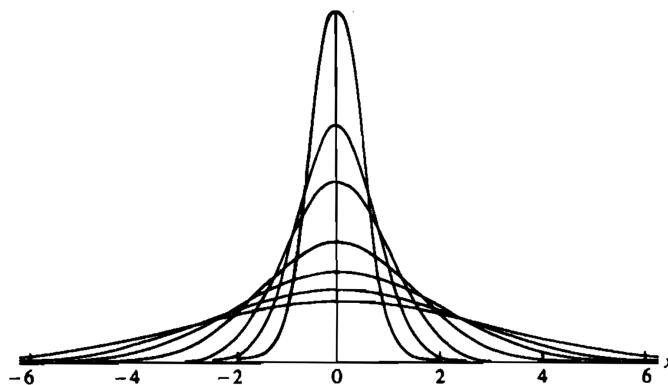


Figure 33.2: Figure 3.5.3. Members of the same scale family

Exponential Families A family of pdfs or pmfs with vector parameter $\boldsymbol{\theta}$ is called an *exponential family* if it can be expressed as

$$f(x|\boldsymbol{\theta}) = h(x)c(\boldsymbol{\theta})\exp\left(\sum_{j=1}^k w_j(\boldsymbol{\theta})t_j(x)\right), \quad x \in S \subset \mathbb{R} \quad (1)$$

where S is not defined in terms of $\boldsymbol{\theta}$, $h(x)$, $c(\boldsymbol{\theta}) \geq 0$ and the functions are just functions of the parameters specified; i.e., $h(x)$ is free of $\boldsymbol{\theta}$, $c(\boldsymbol{\theta})$ is free of x , etc...

Examples:

- One-dimensional: Exponential, Poisson
- Two-dimensional: Gaussian

Exponential family parameterizations are unique except for multiplying constant factors.

Example: Gaussian Let $f(x|\mu, \sigma^2)$ be the $n(\mu, \sigma^2)$ family of pdfs, where $\boldsymbol{\theta} = (\mu, \sigma)$. Then

$$\begin{aligned} f(x|\mu, \sigma^2) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) \exp\left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2}\right) \end{aligned}$$

Thus

$$\begin{aligned} h(x) &= \frac{1}{\sqrt{2\pi}} & c(\mu, \sigma) &= \frac{1}{\sigma} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) \\ w_1(\mu, \sigma) &= -\frac{1}{2\sigma^2} & w_2(\mu, \sigma) &= \frac{\mu}{\sigma^2} \\ t_1(x) &= x^2 & t_2(x) &= x \end{aligned}$$

The parameter space is $(\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$.

Example: Binomial Let $f(x|p)$ be the *binomial*(n, p), $0 < p < 1$ family of pmfs.

$$\begin{aligned} f(x|p) &= \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} (1-p)^n \left[\frac{p}{1-p}\right]^x \\ &= \binom{n}{x} (1-p)^n \exp\left[\log\left(\frac{p}{1-p}\right) x\right] \end{aligned}$$

Thus,

$$\begin{aligned} h(x) &= \binom{n}{x}, \quad x = 0, \dots, n & w_1(p) &= \log\left(\frac{p}{1-p}\right) \\ c(p) &= (1-p)^n, \quad 0 < p < 1 & t_1(x) &= x \end{aligned}$$

Note that this works when p is considered the parameter, while n is fixed. Also, p cannot be 0 or 1. Otherwise, the range changes.

More examples The following distributions belong to Exponential families:

- Continuous: exponential, Gaussian, gamma, beta, χ^2
- Discrete: Poisson, geometric, binomial (fixed # trials), negative binomial (fixed # successes)

The following distributions do not belong to exponential families:

- Continuous: t , F , uniform
For example, $X \sim U(0, \theta)$, $f_X(x) = \theta^{-1} I_{0 < x < \theta}$
- Discrete: uniform, hypergeometric

Theorem If X is a random variable with pdf or pmf of the form 1, then

$$E \left(\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right) = -\frac{\partial}{\partial \theta_j} \log c(\boldsymbol{\theta})$$

$$Var \left(\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right) = -\frac{\partial^2}{\partial \theta_j^2} \log c(\boldsymbol{\theta}) - E \left(\sum_{i=1}^k \frac{\partial^2 w_i(\boldsymbol{\theta})}{\partial \theta_j^2} t_i(X) \right).$$

Although these equations may look formidable, when applied to specific cases they can work out quite nicely. Their advantage is that we can replace integration or summation by differentiation, which is often more straightforward.

Example (Normal exponential family) Let $f(x|\mu, \sigma^2)$ be the $n(\mu, \sigma^2)$ family of pdfs, where $\boldsymbol{\theta} = (\mu, \sigma)$, $-\infty < \mu < \infty, \sigma > 0$. Then

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{\mu^2}{2\sigma^2} \right) \exp \left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} \right)$$

Define

$$\theta_1 = \frac{1}{\sigma^2} > 0, \quad \theta_2 = \frac{\mu}{\sigma^2} \in \mathbb{R}$$

Then

$$f_X(x) = \frac{\sqrt{\theta_1}}{\sqrt{2\pi}} \exp \left(-\frac{\theta_2^2}{2\theta_1} \right) \exp \left(-\theta_1 \frac{x^2}{2} + \theta_2 x \right)$$

and

$$h(x) = 1 \text{ for all } x;$$

$$c(\boldsymbol{\theta}) = c(\theta_1, \theta_2) = \exp \left(-\frac{\theta_2^2}{2\theta_1} \right), \quad (\theta_1, \theta_2) \in (0, \infty) \times \mathbb{R}$$

$$w_1(\boldsymbol{\theta}) = \theta_1 \quad t_1(x) = -x^2/2$$

$$w_2(\boldsymbol{\theta}) = \theta_2 \quad t_2(x) = x$$

Therefore, by the above theorem

$$\begin{aligned} E(X) &= -\frac{\partial}{\partial \theta_2} \log c(\boldsymbol{\theta}) = \frac{\theta_2}{\theta_1} = \mu \\ \text{Var}(X) &= -\frac{\partial^2}{\partial \theta_2^2} \log c(\boldsymbol{\theta}) = -\frac{1}{\theta_1} = \sigma^2 \end{aligned} \tag{2}$$

Joint and Marginal Distributions

In previous lectures, we have discussed probability models and computation of probability for events involving only one random variable. These are called *univariate models*.

In an experimental situation, it would be very unusual to observe only the value of one random variable. For example, in an experiment designed to gain information about some health characteristics of a population of people, the body weights of several people in the population might be measured. These different weights would be observations on different random variables, one for each person measured. Multiple observations could also arise because several physical characteristics were measured on each person. Thus, we need to know how to describe and use probability models that deal with more than one random variable at a time.

Definition: An n -dimensional random vector $\mathbf{X} = (X_1, \dots, X_n)$ is a function from a sample space S into $\mathbb{R}^n$.

- Each coordinate X_i is a random variable.
- The random vector is associated with a probability space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), F)$.
- For each Borel set B ,

$$\Pr\{\mathbf{X} \in B\} = \Pr\{\mathbf{X}^{-1}(B)\} \tag{3}$$

where

$$\mathbf{X}^{-1}(B) = \{w : \mathbf{X}(w) \in B\}$$

Example (Bivariate random variable) A fair coin is flipped 3 times. Define the random vector (X, Y) where X represents the number of heads on the last toss and Y the total number of heads. Then, the probabilities of various outcomes are given in the following table:

Outcome	(x, y)	$\Pr(\text{outcome})$
(H, H, H)	(1, 3)	1/8
(H, T, H), (T, H, H)	(1, 2)	2/8
(H, H, T)	(0, 2)	1/8
(T, T, H)	(1, 1)	1/8
(T, H, T), (H, T, T)	(0, 1)	2/8
(T, T, T)	(0, 0)	1/8

Definition Two random variables X and Y are said to be jointly *discrete* if there is an associated *joint probability mass function*,

$$f_{X,Y}(x, y) = \Pr\{X = x, Y = y\}$$

which sums to 1 over a finite or possibly countable combinations of x and y for which $f_{X,Y}(x, y) > 0$, i.e.,

$$\sum_{x,y} f_{X,Y}(x, y) = 1$$

From this, one can also obtain the marginal pmfs of X and Y as follows:

$$f_X(x) = \Pr(X = x) = \sum_y f_{X,Y}(x, y)$$

$$f_Y(y) = \Pr(Y = y) = \sum_x f_{X,Y}(x, y)$$

Example Back to the fair coin example again. From the definition, we can construct the joint pmf of X and Y :

		Y			
		0	1	2	3
X	0	1/8	1/4	1/8	0
	1	0	1/8	1/4	1/8

The marginal distributions of X and Y are also easy to find. Note: Marginals do not determine joint pmf.

Bivariate cdfs Whether they are discrete or continuous or some combination of the two, we can always define the *joint cdf*. For $n = 2$, the *bivariate cumulative distribution function* is

$$F_{X,Y}(x, y) = \Pr\{X \leq x, Y \leq y\}$$

Properties:

- $F_{X,Y}(x, y) \geq 0$
- $F_{X,Y}(\infty, \infty) = 1$
- $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$
- $F_{X,Y}(-\infty, -\infty) = 0$
- F is non-decreasing and right-continuous in each variable separately.

Joint probabilities All joint probability statements about X and Y can be answered in terms of their joint cdf:

$$\Pr(x_1 < X \leq x_2, y_1 < Y \leq y_2) =$$

$$F_{X,Y}(x_2, y_2) + F_{X,Y}(x_1, y_1) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1)$$

Example

$$\Pr(X > x, Y > y) = 1 - F_X(x) - F_Y(y) + F_{X,Y}(x, y)$$

Note: To ensure that a bivariate function $F(x, y)$ is a proper cdf, it must satisfy all the properties mentioned above and the rectangular property above.

Marginal distributions From $F_{X,Y}$, we can derive the univariate distribution functions for X and Y . These are generally called *marginal distributions*.

$$F_X(x) = \Pr\{X \leq x\} = \Pr\{X \leq x, Y \leq \infty\} = F_{X,Y}(x, \infty)$$

$$F_Y(y) = \Pr\{Y \leq y\} = \Pr\{X < \infty, Y \leq y\} = F_{X,Y}(\infty, y)$$

Note: Although we can obtain $F_X(x)$ and $F_Y(y)$ from the joint cdf, we cannot do the reverse.

Continuous Bivariate RVs The random variables X and Y are said to be *jointly continuous* if there exists a function $f_{X,Y}(x, y)$, such that for any Borel set B of 2-tuples in $\mathbb{R}^2$,

$$\Pr\{(X, Y) \in B\} = \int \int_{(x,y) \in B} f_{X,Y}(x, y) dx dy.$$

The function $f_{X,Y}(x, y)$ is called the *joint probability density function* for X and Y . It follows in this case that

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) dt ds,$$
$$f_{X,Y}(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Properties of the bivariate pdf

- $f_{X,Y}(x, y) \geq 0$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
- $f_{X,Y}(x, y)$ is not a probability, but can be thought of as a relative probability of (X, Y) falling into a small rectangle located at (x, y) :

$$\Pr\{x < X \leq x + dx, y < Y \leq y + dy\} \approx f(x, y) dx dy$$

- The *marginal probability density functions* for X and Y can be obtained as

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Example 1

$$F_{X,Y}(x, y) = xy \quad 0 < x \leq 1, 0 < y \leq 1$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} =$$

$$f_X(x) =$$

$$f_Y(y) =$$

Example 2

$$F_{X,Y}(x, y) = x - x \log \frac{x}{y} \quad 0 < x \leq y \leq 1$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} =$$

$$f_X(x) =$$

$$f_Y(y) =$$

Note: Once we have $f_X(x)$ and $f_Y(y)$, we can obtain $F_X(x)$ and $F_Y(y)$ directly. Double check: $F_X(x) = F_{X,Y}(x, \infty)$.

Conditional Distributions

Conditional Distributions - Discrete Recall if A and B are two events, the probability of A conditional on B is:

$$\Pr(A|B) = \frac{\Pr(A, B)}{\Pr(B)}$$

Defining the events $A = \{Y = y\}$ and $B = \{X = x\}$, it follows that

$$\begin{aligned} \Pr\{Y = y|X = x\} &= \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} \\ &= \frac{f_{X,Y}(x, y)}{f_X(x)} \\ &= f_{Y|X}(y|x) \end{aligned}$$

This is called the *conditional probability mass function* of Y given X .

Example: Discrete Back to the fair coin example. From the joint pmf of X and Y , we can derive all the conditional pmfs:

		Y			
		0	1	2	3
X	0	1/8	1/4	1/8	0
	1	0	1/8	1/4	1/8